

Jump-Model Regimes and Two-Asset Allocation

Data and return-feature construction

- Monthly sample: **1970-01-31 to 2024-11-30**
- Series: equity returns, aggregate bond returns, risk-free return, and macro predictors
- Asset excess return: $r_t^e = r_t - r_t^f$
- Exponentially weighted mean with halflife h :

$$\mu_{t,h} = \frac{\sum_{j \geq 0} 2^{-j/h} r_{t-j}^e}{\sum_{j \geq 0} 2^{-j/h}}$$

- Downside deviation:

$$DD_{t,h} = \sqrt{\frac{\sum_{j \geq 0} 2^{-j/h} \min(r_{t-j}^e, 0)^2}{\sum_{j \geq 0} 2^{-j/h}}}$$

- Log downside deviation:

$$\log DD_{t,h} = \log(DD_{t,h} + \varepsilon)$$

- Equity risk-adjusted feature:

$$\text{Sortino}_{t,h} = \frac{\mu_{t,h}}{DD_{t,h} + \varepsilon}$$

- Bond risk-adjusted feature:

$$\text{Sharpe}_{t,h} = \frac{\mu_{t,h}}{\sigma_{t,h}}$$

Feature families and macro block

- Two monthly feature families with different halflife sets
- Halflives give smooth recursive features with short- and medium-horizon memory
- Shared macro block:

$$(\Delta y_t^{(2Y)}, \text{slope}_t, \Delta \text{slope}_t, \text{VIXtrend}_t, \rho_{t,12m}^{EQ,FI})$$

- These are:
 - 2Y yield change
 - 10Y–2Y slope level
 - 10Y–2Y slope change
 - VIX trend
 - rolling equity–bond correlation

For each asset and month, the jump model observes a transformed feature vector z_t built from the corresponding return-feature family and the shared macro block.

Jump-model regime identification

For each asset, the two-state jump model solves

$$\min_{\{s_t\}, \{\mu_k\}} \sum_{t=1}^T \|z_t - \mu_{s_t}\|^2 + \lambda \sum_{t=2}^T \mathbf{1}(s_t \neq s_{t-1}), \quad K = 2.$$

- $s_t \in \{1, 2\}$: latent state at month t
- μ_k : feature-space center of state k
- λ : persistence penalty; larger λ gives stickier state paths

Raw states are mapped to bull and bear by cumulative excess return:

$$\text{bull} = \arg \max_k \sum_{t: s_t=k} r_t^e, \quad \text{bear} = \arg \min_k \sum_{t: s_t=k} r_t^e.$$

Final choice

$$\lambda^{FI} = 3, \quad \lambda^{EQ} = 10$$

Bond: best annualized Sharpe among non-degenerate candidates with genuine switching.

Equity: best annualized Sharpe and strongest next-state performance before collapse.

Lambda - bond

λ	balance	sw.	$P(00, 01, 10, 11)$	run	bull / bear ex.	next acc.	bal.	acc.	Sharpe	invest
0-0.03	58.4/41.6	117	[.788, .212, .153, .847]	5.585	0.0091/ - 0.0098	0.829	0.819	0.704	62.1%	
0.1	57.8/42.2	113	[.798, .202, .150, .850]	5.781	0.0091/ - 0.0096	0.836	0.827	0.727	61.4%	
0.3	58.0/42.0	107	[.808, .192, .141, .859]	6.102	0.0090/ - 0.0096	0.843	0.834	0.739	61.4%	
1	58.4/41.6	89	[.839, .161, .117, .883]	7.322	0.0087/ - 0.0093	0.874	0.866	0.794	61.9%	
3	59.9/40.1	59	[.890, .110, .076, .924]	10.983	0.0077/ - 0.0086	0.918	0.911	0.859	64.2%	
10	73.6/26.4	24	[.931, .069, .025, .975]	26.360	0.0052/ - 0.0100	0.968	0.954	0.855	77.6%	
30	75.7/24.3	3	[.994, .006, .004, .996]	164.750	0.0033/ - 0.0054	0.997	0.995	0.760	80.4%	
100	71.5/28.5	2	[.995, .005, .002, .998]	219.667	0.0032/ - 0.0039	0.997	0.995	0.760	80.4%	

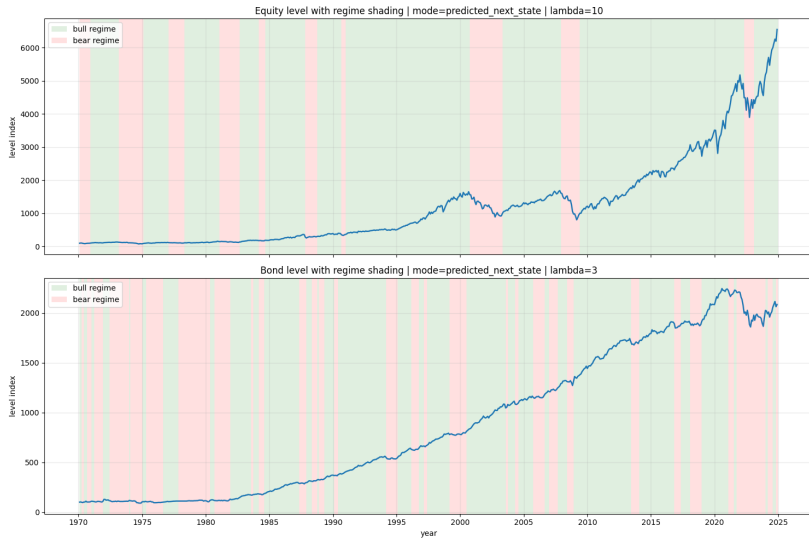
Chosen: $\lambda = 3$. Larger values start collapsing the two-state process; smaller values switch too frequently.

Lambda screening - equity

λ	balance	sw.	$P(00, 01, 10, 11)$	run	bull / bear ex.	next acc.	bal.	acc.	Sharpe	invest
0-0.03	55.8/44.2	131	[.773, .227, .177, .823]	4.992	0.0214/ - 0.0186	0.801	0.798	0.278	0.278	55.8%
0.1	57.4/42.6	119	[.786, .214, .156, .844]	5.492	0.0208/ - 0.0192	0.819	0.815	0.345	0.345	57.3%
0.3	58.1/41.9	111	[.797, .203, .144, .856]	5.884	0.0202/ - 0.0192	0.831	0.827	0.363	0.363	58.1%
1	66.6/33.4	69	[.841, .159, .078, .922]	9.414	0.0168/ - 0.0223	0.895	0.883	0.481	0.481	66.6%
3	69.0/31.0	49	[.877, .123, .053, .947]	13.180	0.0151/ - 0.0216	0.926	0.913	0.576	0.576	69.0%
10	77.8/22.2	19	[.932, .068, .018, .982]	32.950	0.0111/ - 0.0220	0.971	0.959	0.643	0.643	77.8%
30	96.4/3.6	4	[.917, .083, .003, .997]	131.800	0.0033/0.0153	0.994	0.957	0.278	0.278	96.4%
100	99.4/0.6	2	[.750, .250, .002, .998]	219.667	0.0036/0.0281	0.997	0.874	0.296	0.296	99.4%

Chosen: $\lambda = 10$. It maximizes Sharpe before the state process becomes nearly one-sided.

Chosen state paths



Forecasting the final regime states

Final carried-forward targets:

$$\lambda^{FI} = 3, \quad \lambda^{EQ} = 10$$

$$y_{t+1} \in \{0, 1\}, \quad y_{t+1} = 1 \iff \text{bull at } t + 1$$

Rolling design

- training window: 11 years
- validation window: 5 years
- update frequency: every 6 months
- within each update:



- classifier output:
 $p_t = \Pr(y_{t+1} = 1 \mid X_t)$

- trading rule: $\hat{y}_{t+1} = \mathbf{1}\{p_t \geq 0.5\}$

After each 6-month test block, the whole window moves forward by 6 months and the procedure repeats.

Final predictor sets Equity (13):

(logdd_{h1}, logdd_{h4}, mean_{h1}, mean_{h2}, mean_{h4},
sortino_{h1}, sortino_{h2}, sortino_{h4},
ratechg_{2y}, slope, Δslope, vixtrend, $\rho_{12m}^{EQ,FI}$)

Bonds (11):

(mean_{h1}, mean_{h2}, mean_{h4}, sharpe_{h1}, sharpe_{h2}, sharpe_{h4},
ratechg_{2y}, slope, Δslope, vixtrend, $\rho_{12m}^{EQ,FI}$)

Final predictor sets and frozen specification

Model Selection and Configuration:

- **Shallow Trees:** XGBoost grid restricted to $max_depth \in \{2, 3\}$ to prevent overfitting macro noise.
- **Selection Rule:** Hyperparameters optimized via **Validation Sharpe Ratio** (0/1 strategy) rather than raw accuracy.

Equity Predictors (13):

$(\logdd_{h1}, \logdd_{h4}, \text{mean}_{h1}, \text{mean}_{h2}, \text{mean}_{h4},$
 $\text{sortino}_{h1}, \text{sortino}_{h2}, \text{sortino}_{h4},$
 $\text{ratechg}_{2y}, \text{slope}, \Delta\text{slope}, \text{vixtrend}, \rho_{12m}^{EQ, FI})$

Bond Predictors (11):

$(\text{mean}_{h1}, \text{mean}_{h2}, \text{mean}_{h4}, \text{sharpe}_{h1}, \text{sharpe}_{h2}, \text{sharpe}_{h4},$
 $\text{ratechg}_{2y}, \text{slope}, \Delta\text{slope}, \text{vixtrend}, \rho_{12m}^{EQ, FI})$

Final production specification: XGBoost for Bond $\lambda = 3$ and Equity $\lambda = 10$.

Final forecasting results

asset	model	next acc.	bal. acc.	mean ex.	vol	ann. Sharpe
bond	XGBoost	0.8249	0.7788	0.001902	0.009129	0.7218
bond	carry	0.9181	0.9109	0.002559	0.010320	0.8589
equity	XGBoost	0.8777	0.6598	0.005965	0.039481	0.5233
equity	carry	0.9711	0.9592	0.006484	0.034954	0.6426

Differences versus carry benchmark:

$$\Delta \text{Sharpe}^{FI} = -0.1371, \quad \Delta \text{mean ex.}^{FI} = -0.000657$$

$$\Delta \text{Sharpe}^{EQ} = -0.1193, \quad \Delta \text{mean ex.}^{EQ} = -0.000520$$

Takeaway

The carry-forward benchmark remained stronger than the supervised XGBoost forecast for both assets.

correct

Two-asset allocation setup

Common evaluation sample:

1992-01-31 to 2024-10-31

Portfolio net growth:

$$1 + r_{t+1}^{P,\text{net}} = (1 - \text{TC}_t)(1 + r_{t+1}^{P,\text{gross}})$$
$$r_{t+1}^{P,\text{excess}} = r_{t+1}^{P,\text{net}} - r_{t+1}^f$$

Turnover and transaction cost:

$$\text{TO}_t = |w_t^E - \tilde{w}_t^E| + |w_t^B - \tilde{w}_t^B|$$
$$\text{TC}_t = c \cdot \text{TO}_t, \quad c = 0.0005$$

Risk and mean inputs:

Σ_t from exponentially weighted covariance of equity/bond excess returns, halflife 12

$$\mu_t^{\text{MV}} = \text{EWM}_{60}(r^e)$$

For regime-conditioned strategies, each asset mean input is the trailing 11-year regime-conditional average.

Allocation rules

Static benchmark

$$w_t = (0.6, 0.4, 0)$$

Plain equal-weight

$$w_t = (0.5, 0.5, 0)$$

Regime equal-weight

$$w_t = \begin{cases} (0.5, 0.5, 0), & (\hat{y}_t^E, \hat{y}_t^B) = (1, 1), \\ (1, 0, 0), & (\hat{y}_t^E, \hat{y}_t^B) = (1, 0), \\ (0, 1, 0), & (\hat{y}_t^E, \hat{y}_t^B) = (0, 1), \\ (0, 0, 1), & (\hat{y}_t^E, \hat{y}_t^B) = (0, 0). \end{cases}$$

Optimizer constraint set:

$$w_t^E \geq 0, \quad w_t^B \geq 0, \quad w_t^E + w_t^B \leq 1, \quad w_t^C = 1 - w_t^E - w_t^B$$

Grid-search objective, step size 0.0025:

$$\max_{w_t} w_t^\top \mu_t - \gamma_{\text{risk}} w_t^\top \Sigma_t w_t - \gamma_{\text{trade}} c \left(|w_t^E - \tilde{w}_t^E| + |w_t^B - \tilde{w}_t^B| \right)$$

Minimum-variance and mean-variance families

Minimum-variance family

Plain MinVar:

$$\mu_t = (0.001, 0.001)$$

Binary regime MinVar:

$$\mu_t^E = \begin{cases} 0.001, & \hat{y}_t^E = 1 \\ 0, & \hat{y}_t^E = 0 \end{cases}, \quad \mu_t^B = \begin{cases} 0.001, & \hat{y}_t^B = 1 \\ 0, & \hat{y}_t^B = 0 \end{cases}$$

Probability-weighted MinVar:

$$\mu_t^E = 0.001 p_t^E, \quad \mu_t^B = 0.001 p_t^B$$

Reported setting:

$$\gamma_{\text{risk}} = 10, \quad \gamma_{\text{trade}} = 1$$

Mean-variance family

Plain MV:

$$\mu_t = \text{EWM}_{60}(r^e)$$

Binary regime MV:

$$\mu_t^E = \begin{cases} \bar{r}_{t,\text{bull}}^E, & \hat{y}_t^E = 1 \\ \bar{r}_{t,\text{bear}}^E, & \hat{y}_t^E = 0 \end{cases}$$

$$\mu_t^B = \begin{cases} \bar{r}_{t,\text{bull}}^B, & \hat{y}_t^B = 1 \\ \bar{r}_{t,\text{bear}}^B, & \hat{y}_t^B = 0 \end{cases}$$

Probability-weighted MV:

$$\mu_t^E = p_t^E \bar{r}_{t,\text{bull}}^E + (1 - p_t^E) \bar{r}_{t,\text{bear}}^E$$

$$\mu_t^B = p_t^B \bar{r}_{t,\text{bull}}^B + (1 - p_t^B) \bar{r}_{t,\text{bear}}^B$$

Reported settings:

$$\gamma_{\text{risk}} = 5 \text{ (plain MV)}, \quad \gamma_{\text{risk}} = 10 \text{ (regime MV)}, \quad \gamma_{\text{trade}} = 1$$

Reported strategy set

- 60/40
- plain equal-weight
- regime-based equal-weight
- plain minimum-variance
- regime-based minimum-variance with binary forecasts
- plain mean-variance
- regime-based mean-variance with binary forecasts

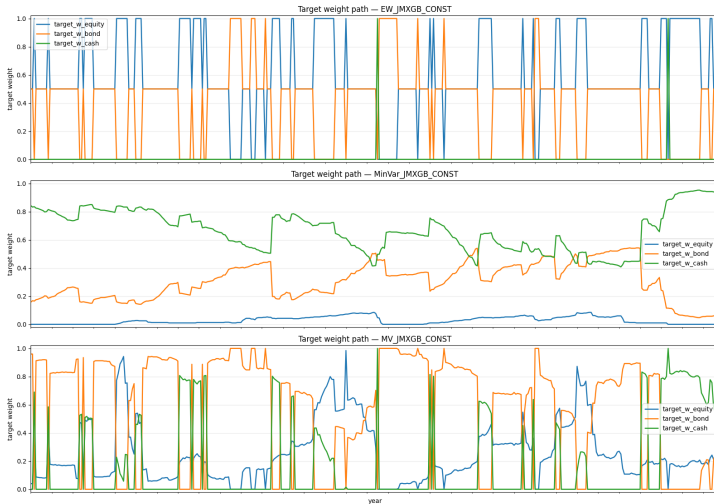
Probability-weighted optimizer variants are robustness checks only.

Allocation results



1. 60_40 is the static benchmark with constant 60% equity and 40% fixed income exposure.
 2. Risk MinVar and plain MV use only historical mean/covariance inputs, with no regime forecast overlay.
 3. CONST strategies use binary next state forecasts for conditional mean inputs. PROB strategies use forecast probabilities only in the mean inputs.

Target weight paths



These panels show how each strategy reallocates across equity, fixed income, and cash over time. They help separate persistent reallocation behavior from one-off episodic timing gains.

Feature-space coupled-state construction

Keep the existing single-asset JM feature blocks and define summary scores

$$S_t^E = \frac{1}{8} \sum_{j=1}^8 \tilde{z}_{t,j}^E, \quad S_t^B = \frac{1}{6} \sum_{j=1}^6 \tilde{z}_{t,j}^B,$$

where all JM inputs are z-scored and equity downside terms enter with flipped sign, so larger score means better conditions.

Define coupled feature-space states

$$J_t^{BB} = \mathbf{1}\{S_t^E < 0, S_t^B < 0\}, \quad J_t^{GG} = \mathbf{1}\{S_t^E > 0, S_t^B > 0\},$$
$$J_t^{OPP} = \mathbf{1}\{\text{sign}(S_t^E) \neq \text{sign}(S_t^B)\}.$$

Empirical frequencies:

$$\Pr(J_t^{BB} = 1) = 0.2595, \quad \Pr(J_t^{GG} = 1) = 0.3035, \quad \Pr(J_t^{OPP} = 1) = 0.4370.$$

Conditional probabilities in same point of time.

Representative conditioning block

Retained macro-financial representatives:

$x_t = (\text{raw_DGS1}, \text{derived_belly_slope}, \text{CORESTICKM159SFRBATL}, \text{VIXCLSx}, \text{mktrf}, \text{bond_10y_mom_12m}, \text{NAPMPRI})$.

Interpretation:

- raw_DGS1: short-rate level
- derived_belly_slope: curve-shape proxy
- CORESTICKM159SFRBATL: sticky core inflation
- VIXCLSx: market stress
- mktrf: equity trend
- bond_10y_mom_12m: bond trend
- NAPMPRI: cyclical / inflation-sensitive activity proxy

Each variable enters standardized:

$$z_{t,j} = \frac{x_{t,j} - \bar{x}_j}{s_j}.$$

Screening equations

For each coupled state $Y_t \in \{J_t^{BB}, J_t^{GG}, J_t^{OPP}\}$, run

$$\Pr(Y_t = 1 \mid z_{t,j}) = \Lambda(\alpha_j + \beta_j z_{t,j}), \quad \Lambda(u) = \frac{e^u}{1 + e^u},$$

and the joint screen

$$\Pr(Y_t = 1 \mid z_t) = \Lambda\left(\alpha + \sum_{j=1}^7 \beta_j z_{t,j}\right).$$

Report:

- univariate slope β_j and p-value
- multivariate slope β_j and p-value
- low/high bucket frequencies

Univariate screen gives direction. Multivariate screen gives retained effect after overlap removal.

Joint-bad state: results

Univariate logit: all seven representatives significant.

Main univariate slopes:

$$\beta_{\text{curve}}^{BB} = -1.3378, \quad p < 10^{-6}, \quad \beta_{\text{bond}}^{BB} = -1.2223, \quad p < 10^{-6},$$

$$\beta_{\text{equity}}^{BB} = -0.9143, \quad p < 10^{-6}, \quad \beta_{\text{rate}}^{BB} = 0.8973, \quad p < 10^{-6},$$

$$\beta_{\text{activity}}^{BB} = 0.8722, \quad p < 10^{-6}, \quad \beta_{\pi}^{BB} = 0.6559, \quad p < 10^{-6}.$$

Multivariate retained drivers:

$$\text{mktrf} : \beta = -1.0845, \quad p < 10^{-6},$$

$$\text{bond_10y_mom_12m} : \beta = -0.8871, \quad p = 0.000002,$$

$$\text{derived_belly_slope} : \beta = -0.8244, \quad p = 0.000027,$$

$$\text{NAPMPRI} : \beta = 0.5136, \quad p = 0.000169.$$

Joint bad is primarily weak equity trend, weak bond trend, adverse curve geometry, and weak cyclical backdrop.

Joint-good state: results

Univariate logit: all seven representatives significant.

Main univariate slopes:

$$\begin{aligned}\beta_{\text{equity}}^{GG} &= 0.7770, \quad p < 10^{-6}, & \beta_{\text{bond}}^{GG} &= 0.7413, \quad p < 10^{-6}, \\ \beta_{\text{stress}}^{GG} &= -0.7807, \quad p < 10^{-6}, & \beta_{\pi}^{GG} &= -0.7635, \quad p < 10^{-6}, \\ \beta_{\text{activity}}^{GG} &= -0.5820, \quad p < 10^{-6}, & \beta_{\text{curve}}^{GG} &= 0.5385, \quad p < 10^{-6}.\end{aligned}$$

Multivariate retained drivers:

$$\begin{aligned}\text{VIXCLSx} : \beta &= -1.1808, \quad p < 10^{-6}, \\ \text{mktrf} : \beta &= 1.0195, \quad p < 10^{-6}, \\ \text{bond_10y_mom_12m} : \beta &= 0.9934, \quad p < 10^{-6}, \\ \text{CORESTICKM159SFRBATL} : \beta &= -0.7957, \quad p = 0.0016, \\ \text{NAPMPRI} : \beta &= -0.3326, \quad p = 0.0128.\end{aligned}$$

Joint good is the mirror-image state: strong market trend, low stress, lower inflation pressure, and healthier macro conditions.

Opposition state: results

Univariate significant terms:

$$\text{VIXCLSx} : \beta = 0.3097, \quad p = 0.0002,$$

$$\text{derived_belly_slope} : \beta = 0.2733, \quad p = 0.0006,$$

$$\text{raw_DGS1} : \beta = -0.2340, \quad p = 0.0038.$$

Multivariate screen:

$$\text{VIXCLSx} : \beta = 0.3201, \quad p = 0.0003$$

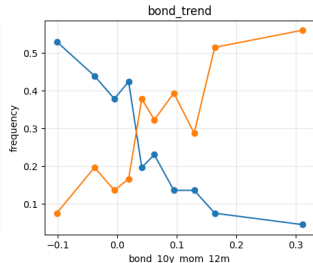
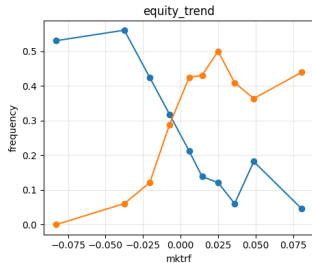
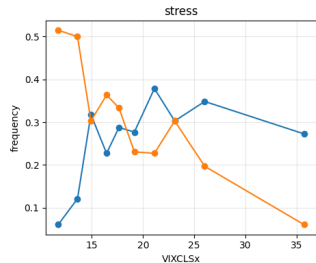
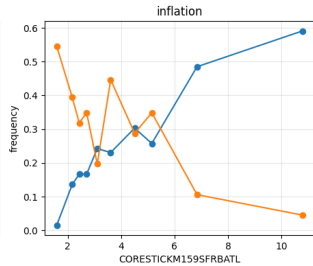
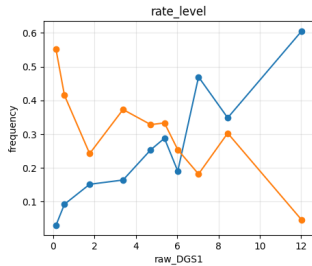
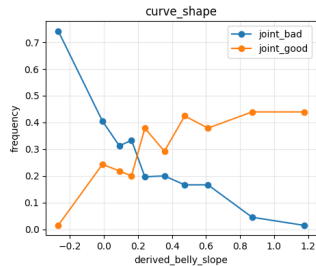
remains the only retained driver, while

$$p_{\text{curve}} = 0.1020, \quad p_{\text{equity}} = 0.1417, \quad p_{\text{activity}} = 0.2581,$$

$$p_{\text{rate}} = 0.2886, \quad p_{\pi} = 0.2988, \quad p_{\text{bond}} = 0.3331.$$

Opposition survives only as a narrow stress-sensitive effect.

Representative signal plots



What the signal plots show

For a retained signal x_t , let $D_1, \dots, D_{10}$ be its deciles. Each panel reports the empirical decile frequencies

$$\Pr(J_t^{BB} = 1 \mid x_t \in D_k) = \frac{1}{|D_k|} \sum_{t \in D_k} J_t^{BB}, \quad \Pr(J_t^{GG} = 1 \mid x_t \in D_k) = \frac{1}{|D_k|} \sum_{t \in D_k} J_t^{GG}, \quad k = 1, \dots, 10.$$

Visible directional patterns:

$$\text{raw_DGS1} \uparrow, \text{CORESTICKM159SFRBATL} \uparrow, \text{VIXCLSx} \uparrow \implies \Pr(J_t^{BB} = 1) \uparrow, \Pr(J_t^{GG} = 1) \downarrow,$$

$$\text{mktrf} \uparrow, \text{bond_10y_mom_12m} \uparrow, \text{derived_belly_slope} \uparrow \implies \Pr(J_t^{BB} = 1) \downarrow, \Pr(J_t^{GG} = 1) \uparrow.$$

So the plots visualize a same-sign good/bad axis:

$$\text{rates / inflation / stress} \uparrow \implies \text{more } BB, \text{ less } GG,$$

$$\text{equity trend / bond trend / curve shape} \uparrow \implies \text{less } BB, \text{ more } GG.$$

From data to allocation

r_t^E = equity return, r_t^B = bond return, r_t^f = risk-free return



$x_t^E = r_t^E - r_t^f$, $x_t^B = r_t^B - r_t^f$



Teacher features from realized excess returns

mean, volatility, downside risk, Sharpe/Sortino, drawdown, recovery, tail-risk features



Teacher models : SJM L2, SJM L1 medoids



Future prediction targets : S_{t+1}^E , S_{t+1}^B

Student feature panel at time t

macro, cross-asset, factor, industry, Treasury, credit, commodity features



Supervised models



$$\hat{p}_t^E = \mathbb{P}(S_{t+H}^E = 1 \mid Z_t), \quad \hat{p}_t^B = \mathbb{P}(S_{t+H}^B = 1 \mid Z_t)$$



Tactical asset allocation (w_t^E, w_t^B, w_t^C)

Teacher feature families

Feature family	Variables	Economic object
SHORT_TAIL_VOL	Short volatility, downside deviation, worst return, lower-tail return, local volatility.	Risk-state teacher.
EWM_RETURN_RISK_DOWNSIDE	EWM mean, Sharpe/Sortino, log downside risk.	Return-risk teacher.
EWM_RETURN_RISK_DRAWDOWN	EWM mean, Sharpe/Sortino, log downside risk, drawdown, drawdown speed, recovery.	Path-aware return-risk teacher.

SHORT_TAIL_VOL: short volatility and tail-risk family

Name			Definition	Why
One-month volatility	annualized	absolute	$vol_t^{1m} = \sqrt{12} x_t $	Current absolute monthly shock.
Three-month volatility	annualized	realized	$vol_t^{3m} = \sqrt{12} \text{sd}(x_t, x_{t-1}, x_{t-2})$	Short realized volatility.
Three-month volatility change			$\Delta vol_t^{3m} = vol_t^{3m} - vol_{t-3}^{3m}$	Acceleration in short volatility.
One-to-three-month volatility ratio			$VR_t = \frac{vol_t^{1m}}{vol_t^{3m}}$	Current shock relative to recent volatility.
Local six-month volatility			$\sigma_{loc,t}^{6m} = \text{sd}(x_t, \dots, x_{t-5})$	Local variability of recent returns.

SHORT_TAIL_VOL: downside and tail variables

Name	Definition	Why
Annualized downside deviation	$D_t^w = \sqrt{12} \sqrt{\frac{1}{w} \sum_{j=0}^{w-1} \min(x_{t-j}, 0)^2}, \quad w \in \{1, 3\}$	Volatility from negative returns only.
Log downside deviation	$\ell D_t^w = \log(1 + D_t^w), \quad w \in \{1, 3\}$	Compressed downside-risk scale.
Worst three-month return	$L_t^{3m} = \min(x_t, x_{t-1}, x_{t-2})$	Largest recent monthly loss.
Six-month 10th percentile return	$Q_{10,t}^{6m} = Q_{0.10}(x_t, \dots, x_{t-5})$	Lower-tail return over the recent window.

EWM_RETURN_RISK_DOWNSIDE: return-risk family

Name	Definition	Why
EWM excess-return mean	$\mu_t^h = \text{EWM}_h(x_t), \quad h \in \{1, 2, 4\}$	Persistent return signal.
EWM excess-return volatility	$\sigma_t^h = \text{EWMsd}_h(x_t), \quad h \in \{1, 2, 4\}$	Persistent volatility signal.
EWM downside risk	$\delta_t^h = \sqrt{\text{EWM}_h(\min(x_t, 0)^2)}, \quad h \in \{1, 2, 4\}$	Persistent downside-risk signal.
Log EWM downside risk	$\ell\delta_t^h = \log(\delta_t^h + 10^{-12}), \quad h \in \{1, 2, 4\}$	Compressed downside-risk measure.
EWM Sortino ratio	$SO_t^h = \frac{\mu_t^h}{\delta_t^h + 10^{-12}}, \quad h \in \{1, 2, 4\}$	Return per unit downside risk.
EWM Sharpe ratio	$SR_t^h = \frac{\mu_t^h}{\sigma_t^h + 10^{-12}}, \quad h \in \{1, 2, 4\}$	Return per unit volatility.

EWM_RETURN_RISK_DRAWDOWN: drawdown variables

Name	Definition	Why
Rolling drawdown	$DD_t^w = \frac{W_t}{\max_{j=0, \dots, w-1} W_{t-j}} - 1, \quad w \in \{3, 6, 12\}$	Distance from rolling peak.
Rolling recovery	$Rec_t^w = \frac{W_t}{\min_{j=0, \dots, w-1} W_{t-j}} - 1, \quad w \in \{3, 6, 12\}$	Recovery from rolling trough.
Drawdown speed	$DDS_t^w = DD_t - DD_{t-w}, \quad w \in \{1, 3, 6\}$	Change in draw-down pressure.
EWM drawdown	$\widetilde{DD}_t^h = \text{EWM}_h(DD_t), \quad h \in \{1, 2, 4\}$	Persistent draw-down level.
EWM recovery	$\widetilde{Rec}_t^h = \text{EWM}_h(Rec_t^{3m}), \quad h \in \{1, 2, 4\}$	Persistent recovery signal.

EWM_RETURN_RISK_DRAWDOWN: final family

Name	Definition	Why
EWM excess-return mean	$\mu_t^h = \text{EWM}_h(x_t), \quad h \in \{1, 2, 4\}$	Persistent return signal.
EWM risk-adjusted return	$SO_t^h = \frac{\mu_t^h}{\delta_t^h + 10^{-12}}, \quad SR_t^h = \frac{\mu_t^h}{\sigma_t^h + 10^{-12}}$	Return relative to downside risk or volatility.
Log EWM downside risk	$\ell\delta_t^h = \log(\delta_t^h + 10^{-12}), \quad h \in \{1, 2, 4\}$	Persistent downside-risk level.
EWM drawdown	$\widetilde{DD}_t^h = \text{EWM}_h(DD_t), \quad h \in \{1, 4\}$	Persistent draw-down state.
EWM drawdown speed	$\Delta \widetilde{DD}_t^{h2} = \widetilde{DD}_t^{h2} - \widetilde{DD}_{t-1}^{h2}$	Recent change in persistent draw-down.
EWM recovery	$\widetilde{Rec}_t^{h4} = \text{EWM}_4(Rec_t^{3m})$	Persistent recovery from troughs.

Teacher fitting scheme

For each asset, feature family, and model specification, the regime label at month t is estimated using only observations available up to month t .

$$X_t \in \mathbb{R}^p, \quad \mathcal{I}_t = \{s \leq t : X_s \text{ observed}\}.$$

for $t = 1, \dots, T$:

if $|\mathcal{I}_t| < 120$: no label

else:

 fit the selected teacher model on $\{X_s : s \in \mathcal{I}_t\}$

 extract the current raw two-state label $\hat{S}_t^{raw}$

 order the two raw states by the feature-family rule

 store the mapped state $\hat{S}_t \in \{0, 1\}$.

Teacher model 1: SJM L2

Two-state jump model

$$S_t \in \{0, 1\}, \quad X_t \in \mathbb{R}^p$$

State centers

$$c_0, c_1 \in \mathbb{R}^p$$

Within-state L2 loss

$$L_{t,k} = \|X_t - c_k\|_2^2 = \sum_{j=1}^p (X_{t,j} - c_{k,j})^2$$

Jump penalty

$$J(S_{1:T}) = \sum_{t=2}^T \mathbf{1}\{S_t \neq S_{t-1}\}$$

Objective

$$\min_{S_{1:T}, c_0, c_1} \sum_{t=1}^T \|X_t - c_{S_t}\|_2^2 + \lambda \sum_{t=2}^T \mathbf{1}\{S_t \neq S_{t-1}\}$$

Center update

$$c_k = \frac{1}{|\{t : S_t = k\}|} \sum_{t: S_t = k} X_t$$

Grid

$$\lambda \in \{0, 0.05, 0.10, 0.25, 0.50, 1, 2, 4, 8, 16\}$$

SJM L2: selected lambda paths

Each row summarizes the fixed-lambda SJM L2 paths that entered the all-teacher majority vote for the corresponding asset-feature family..

Target	lambda	Good/bad share	Sw.	Run	SR good	SR bad	Feature set family	Good/bad metric
Equity downside	8, 16	34–36 / 64–66	43–64	7.3–10.8	1.71–2.65	-0.02 to 0.14	Downside	0.025–0.041 / 0.129–0.130
Equity drawdown	8, 16	41–43 / 57–59	52–62	7.5–9.0	1.72–2.47	-0.18 to -0.01	Drawdown	-0.150 to -0.140 / -0.222 to -0.211
Bond downside	0, 0.05, 0.1, 0.25, 0.5, 1, 2	43–45 / 55–57	69–81	5.8–6.8	3.20–3.56	-0.78 to -0.55	Downside	0.006–0.007 / 0.036
Bond drawdown	2, 4	48 / 52	65–75	6.2–7.2	2.55–2.79	-0.70 to -0.56	Drawdown	-0.083 / -0.081

Teacher model 2: SJM L1 medoids

State variable

$$S_t \in \{0, 1\}, \quad X_t \in \mathbb{R}^p.$$

State representatives

$$m_0, m_1 \in \{X_1, \dots, X_T\}.$$

The representatives are medoids, meaning they are observed months from the sample, not arithmetic averages.

L1 assignment loss

$$L_{t,k} = \|X_t - m_k\|_1 = \sum_{j=1}^p |X_{t,j} - m_{k,j}|.$$

Jump penalty

$$J(S_{1:T}) = \sum_{t=2}^T \mathbf{1}\{S_t \neq S_{t-1}\}.$$

Optimization problem

$$\min_{S_{1:T}, m_0, m_1} \sum_{t=1}^T \|X_t - m_{S_t}\|_1 + \lambda \sum_{t=2}^T \mathbf{1}\{S_t \neq S_{t-1}\}.$$

Medoid update

$$m_k = \arg \min_{X_i: S_i=k} \sum_{t: S_t=k} \|X_t - X_i\|_1.$$

Path update

$$S_{1:T} = \arg \min_{S_{1:T}} \sum_{t=1}^T L_{t,S_t} + \lambda \sum_{t=2}^T \mathbf{1}\{S_t \neq S_{t-1}\}.$$

Grid

$$\lambda \in \{0, 0.05, 0.10, 0.25, 0.50, 1, 2, 4, 8, 16\}$$

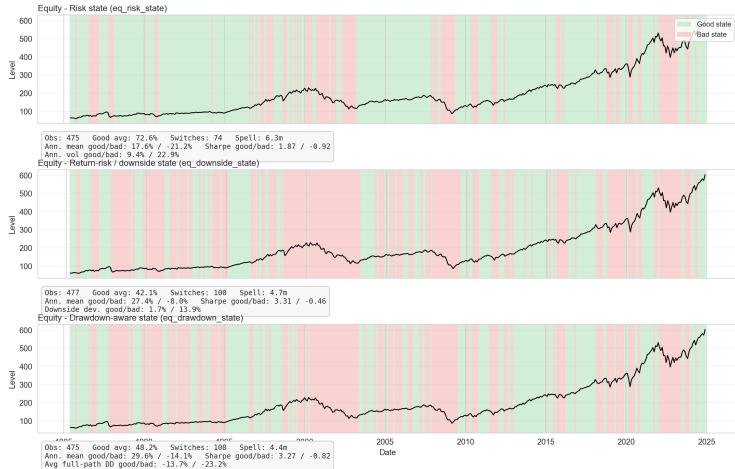
SJM L1 medoids: selected lambda paths

Each row summarizes the fixed-lambda SJM L1-medoid paths that entered the all-teacher majority vote for the corresponding asset–feature family..

Target	lambda	Good/bad share	Sw.	Run	SR good	SR bad	Feature set family	Good/bad metric
Equity risk	2, 4	61–63 / 37–39	52–62	7.5–9.0	1.43–2.01	-0.45 to -0.25	Vol.	0.087–0.099 / 0.209–0.211
Bond risk	0, 0.05, 0.1, 0.25, 0.5	57–61 / 39–43	89–118	4.0–5.3	2.45–2.68	-1.15 to -0.76	Vol.	0.030–0.031 / 0.049–0.050
Bond drawdown	2	48 / 52	71	6.6	2.53	-0.59	Drawdown	-0.089 / -0.075

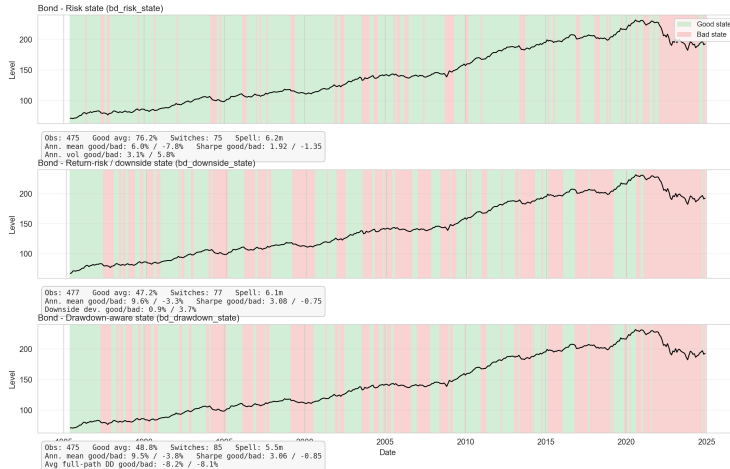
Majority-vote labels: Equity

Majority-vote teacher labels over equity level



Majority-vote labels: Bond

Majority-vote teacher labels over bond level



Student feature groups

Group	Inputs	Why
Macro / policy / cycle	Policy-rate change, yield-curve changes, core inflation, CFNAI, sentiment	Captures monetary policy, growth, inflation, and business-cycle pressure.
Market stress / financial conditions	Credit-spread change, VIX change, realized-volatility change, NFCI, ANFCI	Measures tightening financial conditions, risk aversion, and broad market stress.
Equity	Fama–French factors, size/value portfolios, CRSP 12-industry returns	Which parts of the equity market had stronger or weaker recent returns.
Market-structure summaries	Return dispersion, average correlation, first principal-component share, and changes in correlations	These variables describe how returns inside a block behave together. When dispersion, correlations, or concentration change, the market is no longer behaving like before. This can help the student model detect transitions into different equity or bond states.
Rates / Treasury structure	Treasury maturity-bucket returns, long–short return spread, belly–barbell spread	Captures duration regime, curve positioning, and internal bond-market structure.
Credit / commodities	Investment-grade bond returns, high-yield bond returns, high-yield minus investment-grade return, commodity return, commodity volatility, commodity Sharpe	Credit variables measure safer corporate bonds versus riskier corporate bonds. Commodity variables measure raw-material price pressure and commodity-market risk.
Cross-asset dependence	Equity–bond, equity–Treasury, equity–high yield β , bond–investment grade β , equity–commodity correlation changes	Captures diversification breakdowns and changing dependence across major asset classes.

Student models: families, rolling fitting, and seed averaging

Logistic regression

- Binary classifier for each target
 $Y_t^j \in \{0, 1\}$
- L2 penalty
- $C \in \{0.03, 0.05, 0.10, 0.25, 0.50, 1.00\}$
- Balanced class weights
- Median imputation
- Feature standardization

XGBoost

- Boosted trees with logistic objective
- Max depth $d \in \{1, 2\}$
- Learning rate $\eta \in \{0.03, 0.05\}$
- 200 trees
- $\text{reg_lambda} \in \{10, 30\}$
- $\text{subsample} = 0.85$
- $\text{colsample_bytree} = 0.85$
- $\text{min_child_weight} = 5$
- $\text{scale_pos_weight} = N_0/N_1$
- Median imputation

Random forest

- 500 trees
- Bootstrap aggregation
- $m_{\text{try}} = \sqrt{p}$
- Grid:
 $(d, \ell) \in \{(2, 10), (2, 20), (3, 10), (3, 20)\}$
- d : max depth, ℓ : min leaf size
- Balanced subsample class weights
- Median imputation

Rolling out-of-sample procedure and seed averaging

For each test month t :

- training sample - expanding,
- validation sample = previous 36 months,
- test sample = month t ,
- select the best specification on validation data by lexicographic ranking: AUC – discrimination, balanced accuracy – class balance, Brier – calibration error, net Sharpe – timing payoff, drawdown improvement – tail loss, turnover – trading stability; for risk targets, volatility reduction – risk control is ranked before net Sharpe; then refit on training + validation,
- predict $\hat{p}_t^j = \mathbb{P}(Y_t^j = 1 \mid Z_t)$.

Minimum training length: 96 months. Each selected specification is run with seeds 101, 111, 121, and the final probability is the average across the three seeds.

Out-of-sample student prediction performance

2000-07 to 2024-10, 292 OOS months per target and model family.

Equity targets						Bond targets					
Target	Fam.	AUC	BAcc	Brier	SR	Target	Fam.	AUC	BAcc	Brier	SR
Risk	LOGIT	0.874	0.790	0.140	0.021	Risk	LOGIT	0.770	0.723	0.196	0.130
Risk	XGB	0.837	0.788	0.159	-0.026	Risk	XGB	0.707	0.636	0.221	0.357
Risk	RF	0.850	0.776	0.160	0.051	Risk	RF	0.747	0.708	0.196	0.068
Downside	LOGIT	0.822	0.774	0.170	0.583	Downside	LOGIT	0.840	0.781	0.159	0.429
Downside	XGB	0.809	0.726	0.176	0.503	Downside	XGB	0.820	0.755	0.170	0.475
Downside	RF	0.781	0.731	0.186	0.605	Downside	RF	0.821	0.708	0.173	0.316
Drawdown	LOGIT	0.822	0.786	0.168	0.592	Drawdown	LOGIT	0.830	0.774	0.166	0.546
Drawdown	XGB	0.823	0.749	0.170	0.621	Drawdown	XGB	0.778	0.716	0.190	0.404
Drawdown	RF	0.782	0.713	0.189	0.561	Drawdown	RF	0.791	0.700	0.186	0.390

$$\text{AUC} = \frac{1}{N_1 N_0} \sum_{i:y_i=1} \sum_{j:y_j=0} \mathbf{1}\{\hat{p}_i > \hat{p}_j\}, \quad \text{BAcc} = \frac{1}{2} \left(\frac{TP}{TP + FN} + \frac{TN}{TN + FP} \right), \quad \text{Brier} = \frac{1}{N} \sum_{i=1}^N (\hat{p}_i - y_i)^2, y_i \in \{0, 1\}$$

SR = timing-rule Sharpe.

Methodology: from student probabilities to tactical allocation

Inputs. For each model family $f \in \{\text{LOGIT}, \text{XGB}, \text{RF}\}$ and feature set $g \in \{\text{risk}, \text{downside}, \text{drawdown}\}$, the student model gives

$$\hat{p}_t^{E,g,f} = \mathbb{P}(S_{t+1}^E = 1 \mid Z_t), \quad \hat{p}_t^{B,g,f} = \mathbb{P}(S_{t+1}^B = 1 \mid Z_t).$$

This gives $3 \times 3 = 9$ tactical allocation strategies.

1. Forecast moments from predicted states.

At each month t , state-conditional means and variances are estimated with expanding past data only. They are shrunk toward pre-OOS unconditional moments with fixed effective sample size 36:

$$\omega_{s,t}^a = \frac{n_{s,t}^a}{n_{s,t}^a + 36}.$$

The predicted probability mixes bad-state and good-state moments:

$$\hat{\mu}_t^a = (1 - \hat{p}_t^a) \hat{\mu}_{0,t}^a + \hat{p}_t^a \hat{\mu}_{1,t}^a.$$

The same mixture idea is used for variance. Equity–bond covariance uses the forecast volatilities and a shrunk expanding correlation. This gives $\hat{\mu}_t$ and $\hat{\Sigma}_t$.

2. Optimize portfolio weights.

Weights are long-only with cash allowed:

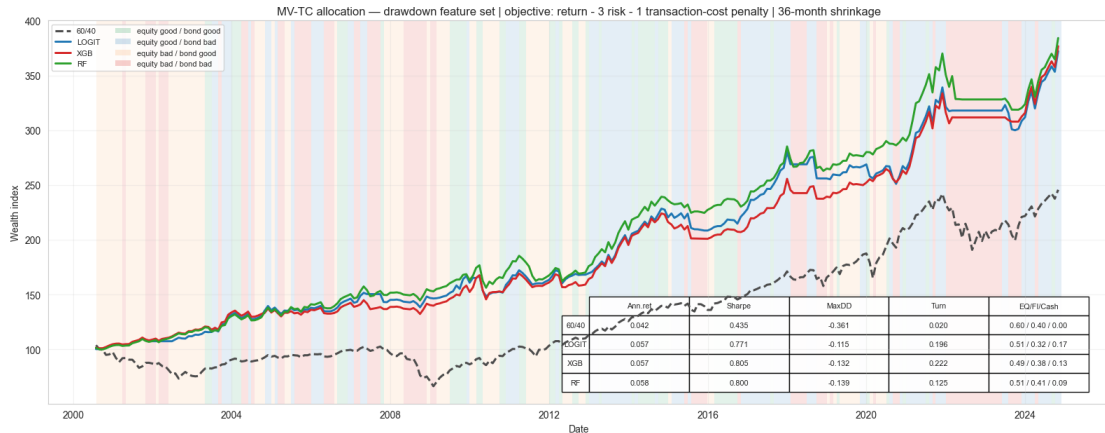
$$w_t^E, w_t^B \in \{0, 0.01, \dots, 1\}, \quad w_t^E + w_t^B \leq 1, \quad w_t^C = 1 - w_t^E - w_t^B.$$

The grid-search objective is

$$\max_{w_t} \left\{ w_t^\top \hat{\mu}_t - 3 w_t^\top \hat{\Sigma}_t w_t - 0.0005(|w_t^E - w_{t-1}^E| + |w_t^B - w_{t-1}^B|) \right\}.$$

3. Evaluate. Realized net returns use the selected weights and the same transaction-cost rule. Benchmark: monthly rebalanced 60/40.

MV-TC allocation: Drawdown feature set



MV-TC allocation: Risk feature set

